

# VALOX<sup>TM</sup> FR RESINS 553

REGION ASIA

## DESCRIPTION

VALOX 553 is a 30% glass filled flame retardant Polybutylene Terephthalate/Polycarbonate (PBT/PC) injection moldable grade. It has excellent chemical resistance and a UL94V0@0.86mm and 5VA@2.3mm flame rating. This is a good candidate for applications needing reduced shrinkage/warpage including appliance handles, spotlights, electric motors, and pump housings

## TYPICAL PROPERTY VALUES

Revision 20190925

| PROPERTIES                                   | TYPICAL VALUES | UNITS              | TEST METHODS |
|----------------------------------------------|----------------|--------------------|--------------|
| <b>MECHANICAL</b>                            |                |                    |              |
| Tensile Stress, brk, Type I, 5 mm/min        | 110            | MPa                | ASTM D 638   |
| Flexural Stress, brk, 1.3 mm/min, 50 mm span | 179            | MPa                | ASTM D 790   |
| Flexural Modulus, 1.3 mm/min, 50 mm span     | 9400           | MPa                | ASTM D 790   |
| Hardness, Rockwell R                         | 118            | -                  | ASTM D 785   |
| <b>IMPACT</b>                                |                |                    |              |
| Izod Impact, unnotched, 23°C                 | 640            | J/m                | ASTM D 4812  |
| Izod Impact, notched, 23°C                   | 85             | J/m                | ASTM D 256   |
| <b>THERMAL</b>                               |                |                    |              |
| HDT, 0.45 MPa, 6.4 mm, unannealed            | 204            | °C                 | ASTM D 648   |
| HDT, 1.82 MPa, 6.4 mm, unannealed            | 160            | °C                 | ASTM D 648   |
| HDT, 1.82 MPa, 3.2mm, unannealed             | 138            | °C                 | ASTM D 648   |
| CTE, -40°C to 40°C, flow                     | 2.3E-05        | 1/°C               | ASTM E 831   |
| CTE, 60°C to 138°C, flow                     | 2.16E-05       | 1/°C               | ASTM E 831   |
| Relative Temp Index, Elec                    | 125            | °C                 | UL 746B      |
| Relative Temp Index, Mech w/impact           | 110            | °C                 | UL 746B      |
| Relative Temp Index, Mech w/o impact         | 125            | °C                 | UL 746B      |
| <b>PHYSICAL</b>                              |                |                    |              |
| Specific Gravity                             | 1.59           | -                  | ASTM D 792   |
| Specific Volume                              | 0.64           | cm <sup>3</sup> /g | ASTM D 792   |
| Water Absorption, 24 hours                   | 0.07           | %                  | ASTM D 570   |
| Moisture Absorption, 50% RH, 24 hrs          | 0.07           | %                  | ASTM D 570   |
| Mold Shrinkage, flow, 1.5-3.2 mm             | 0.3 – 0.5      | %                  | SABIC method |
| Mold Shrinkage, flow, 3.2-4.6 mm             | 0.5 – 0.8      | %                  | SABIC method |
| Mold Shrinkage, xflow, 1.5-3.2 mm            | 0.4 – 0.6      | %                  | SABIC method |
| Mold Shrinkage, xflow, 3.2-4.6 mm            | 0.6 – 0.9      | %                  | SABIC method |
| <b>ELECTRICAL</b>                            |                |                    |              |
| Volume Resistivity                           | 4.3E+16        | Ohm-cm             | ASTM D 257   |
| Dielectric Strength, in air, 3.2 mm          | 18.9           | kV/mm              | ASTM D 149   |
| Dielectric Strength, in oil, 1.6 mm          | 25             | kV/mm              | ASTM D 149   |
| Relative Permittivity, 100 Hz                | 3.8            | -                  | ASTM D 150   |
| Relative Permittivity, 1 MHz                 | 3.7            | -                  | ASTM D 150   |
| Dissipation Factor, 100 Hz                   | 0.002          | -                  | ASTM D 150   |
| Dissipation Factor, 1 MHz                    | 0.02           | -                  | ASTM D 150   |

| PROPERTIES                               | TYPICAL VALUES                 | UNITS    | TEST METHODS |
|------------------------------------------|--------------------------------|----------|--------------|
| Arc Resistance, Tungsten {PLC}           | 6                              | PLC Code | ASTM D 495   |
| Hot Wire Ignition {PLC}                  | 1                              | PLC Code | UL 746A      |
| High Voltage Arc Track Rate {PLC}        | 3                              | PLC Code | UL 746A      |
| High Ampere Arc Ign, surface {PLC}       | 3                              | PLC Code | UL 746A      |
| Comparative Tracking Index (UL) {PLC}    | 3                              | PLC Code | UL 746A      |
| FLAME CHARACTERISTICS                    |                                |          |              |
| UL Yellow Card Link                      | <a href="#">E207780-643582</a> | -        | -            |
| UL Yellow Card Link 2                    | <a href="#">E45587-236818</a>  | -        | -            |
| UL Recognized, 94V-0 Flame Class Rating  | 0.86                           | mm       | UL 94        |
| UL Recognized, 94-5VA Flame Class Rating | 2.3                            | mm       | UL 94        |
| Oxygen Index (LOI)                       | 37.1                           | %        | ASTM D 2863  |
| UV-light, water exposure/immersion       | F1                             | -        | UL 746C      |
| INJECTION MOLDING                        |                                |          |              |
| Drying Temperature                       | 120                            | °C       |              |
| Drying Time                              | 3 – 4                          | hrs      |              |
| Drying Time (Cumulative)                 | 12                             | hrs      |              |
| Maximum Moisture Content                 | 0.02                           | %        |              |
| Melt Temperature                         | 250 – 265                      | °C       |              |
| Nozzle Temperature                       | 245 – 260                      | °C       |              |
| Front - Zone 3 Temperature               | 250 – 265                      | °C       |              |
| Middle - Zone 2 Temperature              | 245 – 260                      | °C       |              |
| Rear - Zone 1 Temperature                | 240 – 255                      | °C       |              |
| Mold Temperature                         | 65 – 90                        | °C       |              |
| Back Pressure                            | 0.3 – 0.7                      | MPa      |              |
| Screw Speed                              | 50 – 80                        | rpm      |              |
| Shot to Cylinder Size                    | 40 – 80                        | %        |              |
| Vent Depth                               | 0.025 – 0.038                  | mm       |              |

## DISCLAIMER

Any sale by SABIC, its subsidiaries and affiliates (each a "seller"), is made exclusively under seller's standard conditions of sale (available upon request) unless agreed otherwise in writing and signed on behalf of the seller. While the information contained herein is given in good faith, SELLER MAKES NO WARRANTY, EXPRESS OR IMPLIED, INCLUDING MERCHANTABILITY AND NONINFRINGEMENT OF INTELLECTUAL PROPERTY, NOR ASSUMES ANY LIABILITY, DIRECT OR INDIRECT, WITH RESPECT TO THE PERFORMANCE, SUITABILITY OR FITNESS FOR INTENDED USE OR PURPOSE OF THESE PRODUCTS IN ANY APPLICATION. Each customer must determine the suitability of seller materials for the customer's particular use through appropriate testing and analysis. No statement by seller concerning a possible use of any product, service or design is intended, or should be construed, to grant any license under any patent or other intellectual property right.